

Data Cleaning Methodology

Data cleaning was performed in Google Sheets before the cleaned dataset was imported into BigQuery for analysis. The following steps were taken:

1. Initial Data Checks

- Verified that there were no blank cells in any columns
- Trimmed unnecessary whitespace and double spaces
- Checked for duplicate games
 - 2 duplicate records were identified and removed
- Checked for any missing or placeholder values
 - N/A values were identified in the Year and Publisher columns
 - “Unknown” values were identified in the Publisher column
 - Publisher information was not filled in because this information was not relevant to the planned analysis
 - Missing Year values were filled using each game’s initial release year

2. Verification of Game Records

- Identified 4 games that were not actually released on their respective platforms and removed these records from the dataset
- After gathering supplemental platform release-year data, identified 4 games with incorrect release years in the original dataset and corrected their Year values

When release dates differed by region, the earliest release date was selected to represent the game’s initial release. This approach was consistent with the organization of the original dataset.

3. Review of Regional and Duplicate Sales Records

Some values in the Name column contained sales-related annotations, such as “Japan sales” or “Other region sales”, that did not appear to be part of the game’s actual title. All entries containing “sales” in the title were reviewed.

For games with multiple records that clearly represented the same game:

- Verified that the information in the remaining columns was identical, otherwise evaluated conflicts in fields such as Genre, Publisher, or Year
- When conflicts were identified, verified the appropriate value using available information and reputable sources
- Combined compatible regional records into a single record when appropriate
- Removed the original records that were replaced by a merged record

For games with multiple records containing conflicting sales information:

- Verified that the records referred to the same game
- Retained the most appropriate record based on the available information
- Removed the secondary record

Sales-related annotations were removed from any games that contained sales-related information in the title but did not have another corresponding record.

4. Recalculation of Global Sales and Rankings

- Recalculated the Global_Sales column by summing the regional sales columns after identifying incorrect values in the original Global_Sales column
- Recalculated the Rank column based on the recalculated Global_Sales values

5. Documentation and Audit Trail

All removed records were included in the vgsales_removed_entries document along with the reason for their removal.

Records that were merged, or had had their Year or Title values updated were documented in the Record Type column. Sources that were used to update information were documented in the Year Source column. The following Record Types were used:

- “New Merged Record” for newly created merged records
- “Original Data - Standardized Title” for records that had updated Title values
- “Original Data - Year Corrected” for records that had updated Year values